

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. **Design a low-latency communication architecture for edge-assisted smart city services. (BL6)**
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)

19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I C Dilipkumar (192412133) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C Dilipkumar

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

1. Design Task:

Design a low-latency communication architecture for edge-assisted smart city services in which computationally intensive and delay-sensitive applications are processed at edge computing nodes located close to end users. The architecture must support low communication latency, efficient resource allocation, service differentiation and reliable communication for smart-city applications.

- Smart-city devices communicate through a 5G wireless access network.
- Edge computing nodes are deployed close to roadside units/base stations.
- Three representative services are considered: **autonomous traffic control, video surveillance and environmental monitoring**.
- 5G radio resources are dynamically allocated according to service priority and traffic demand.
- Edge servers process application tasks locally to reduce cloud round-trip delay.
- Communication and edge-processing latency are evaluated separately.
- MATLAB verification using SNR-dependent throughput, transmission latency and edge-processing calculations.

2. Design Parameters:

Parameter	Designed Value	Purpose
Carrier frequency	3.5 GHz	Sub-6 GHz next-generation wireless operation
System bandwidth	100 MHz	Shared physical spectrum
MIMO	2×2	Spatial multiplexing / diversity
Transmit power	23 dBm	Simulation reference
Noise power	−94 dBm	Receiver noise reference
Number of users	30	Multi-user virtualization test
Virtual slices	3	Service differentiation
Resource blocks	273	Approximate 100 MHz NR-style allocation at 30 kHz SCS
SNR range	−5 to 25 dB	Channel-quality sweep
Implementation efficiency	0.75	Accounts for overhead and practical losses
Allocation method	Weighted QoS scheduling	Fair and service-aware sharing

3. Required Design Formulas:

1 SNR

$$\text{SNR(dB)} = P_r(\text{dBm}) - P_n(\text{dBm})$$
$$\text{SNR(linear)} = 10^{(\text{SNR(dB)}/10)}$$

2 Spectral Efficiency

$$\eta = \log_2(1 + \text{SNR})$$

For practical simulation, η is limited to a maximum of 6 bit/s/Hz.

3 Slice Throughput

$$C_i = B_i \times \eta \times \alpha$$

where B_i is slice bandwidth in MHz, η is spectral efficiency in bit/s/Hz and $\alpha = 0.75$ is the implementation-efficiency factor. With MHz units, C_i is obtained in Mbps.

4 Resource Allocation

$$R_i = \max(R_{i,\min}, \text{round}(w_i \times R_{\text{available}}))$$

subject to $\sum R_i \leq R_{\text{available}}$.

5 QoS Verification

Throughput condition: $C_i \geq C_{i,\min}$

Latency condition: $L_i \leq L_{i,\max}$

A slice is considered successfully provisioned when its resource reservation and QoS constraints are satisfied.

4. Worked Design Calculation:

At an SNR of 15 dB:

$$\text{SNR} = 10^{(15/10)} = 31.62$$

$$\eta = \log_2(1 + 31.62) \approx 5.03 \text{ bit/s/Hz}$$

Using $\alpha = 0.75$:

$$\text{eMBB: } C = 50 \times 5.03 \times 0.75 \approx 188.6 \text{ Mbps}$$

$$\text{URLLC: } C = 30 \times 5.03 \times 0.75 \approx 113.2 \text{ Mbps}$$

$$\text{mMTC: } C = 20 \times 5.03 \times 0.75 \approx 75.5 \text{ Mbps}$$

These are model-based capacities used to verify the design; actual 5G throughput depends on modulation, coding, MIMO rank, scheduling, control overhead and channel conditions.

5. MATLAB Simulation Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
%% =====  
% LOW-LATENCY COMMUNICATION ARCHITECTURE  
% FOR EDGE-ASSISTED SMART CITY SERVICES  
% =====
```

```
%% 1. System Parameters
```

```
fc_GHz = 3.5;          % Carrier frequency (GHz)  
BW_MHz = 100;          % System bandwidth (MHz)  
N_RB = 273;           % Number of resource blocks
```

```
Nt = 2;                % Transmit antennas  
Nr = 2;                % Receive antennas
```

```

Ptx_dBm = 23;          % Transmit power (dBm)
Noise_dBm = -94;       % Noise power (dBm)

Nusers = 30;           % Number of smart-city devices
Nedge = 3;             % Number of edge nodes

etaPractical = 0.75;    % Implementation efficiency
edgeEfficiency = 0.90;  % Edge processing efficiency

```

```

%% 2. Smart City Services

```

```

serviceName = {'Traffic Control', ...
               'Video Surveillance', ...
               'Environmental Monitoring'};

```

```

% Bandwidth allocation (MHz)
serviceBW = [50 30 20];

```

```

% Priority weights
weight = [0.50 0.30 0.20];

```

```

% Minimum throughput requirements (Mbps)
targetRate = [40 20 5];

```

```

% Maximum latency requirements (ms)
maxLatency = [10 20 100];

```

```

%% 3. Resource Block Allocation

```

```

RB = round(weight * N_RB);

```

```

% Correct the final value so total RB = 273
RB(end) = N_RB - sum(RB(1:end-1));

```

```

fprintf('=====\n');
fprintf('RESOURCE BLOCK ALLOCATION\n');
fprintf('=====\n');

```

```

for s = 1:3
    fprintf('%s = %d RB\n', serviceName{s}, RB(s));
end

```

```

fprintf('Total RB = %d RB\n\n', sum(RB));

```

```

%% 4. SNR Sweep

```

```

SNR_dB = -5:5:25;

```

```

% Convert SNR from dB to linear
SNR = 10.^(SNR_dB/10);

```

```

%% 5. Spectral Efficiency

```

```

eta = log2(1 + SNR);

```

```

% Practical maximum spectral efficiency
eta = min(eta, 6);

%% 6. Throughput Calculation

throughput = zeros(length(SNR_dB), 3);

for k = 1:length(SNR_dB)

    for s = 1:3

        % Throughput in Mbps
        throughput(k,s) = ...
            serviceBW(s) * eta(k) * etaPractical;

    end

end

%% 7. Display Throughput at 15 dB

idx = find(SNR_dB == 15);

fprintf('=====\n');
fprintf('THROUGHPUT AT 15 dB SNR\n');
fprintf('=====\n');

for s = 1:3

    fprintf('%s = %.2f Mbps\n', ...
        serviceName{s}, throughput(idx,s));

end

%% 8. Packet Size for Each Service

% Packet size in kB
packetSize_kB = [100 250 50];

% Convert kB to bits
packetBits = packetSize_kB * 8 * 1000;

%% 9. Transmission Latency

txLatency = zeros(1,3);

for s = 1:3

    % Convert Mbps to bps
    rate_bps = throughput(idx,s) * 1e6;

    % Transmission latency in milliseconds
    txLatency(s) = ...
        (packetBits(s) / rate_bps) * 1000;

```

```

end

%% 10. Edge Processing Parameters

% CPU cycles required by each application
CPUcycles = [1e8 2e8 5e7];

% Edge CPU frequency in GHz
edgeCPU_GHz = [3 3 3];

%% 11. Edge Processing Latency

edgeLatency = zeros(1,3);

for s = 1:3

    edgeLatency(s) = ...
        CPUcycles(s) / ...
        (edgeCPU_GHz(s) * 1e9 * edgeEfficiency) * 1000;

end

%% 12. Queueing / Scheduling Latency

queueLatency = [2 3 5]; % milliseconds

%% 13. Total Latency

totalLatency = txLatency + ...
    edgeLatency + ...
    queueLatency;

%% 14. Display Latency Results

fprintf('\n');
fprintf('=====\n');
fprintf('LATENCY RESULTS AT 15 dB SNR\n');
fprintf('=====\n');

for s = 1:3

    fprintf('\n%s\n', serviceName{s});

    fprintf('Transmission Latency = %.2f ms\n', ...
        txLatency(s));

    fprintf('Edge Processing Latency = %.2f ms\n', ...
        edgeLatency(s));

    fprintf('Queueing Latency = %.2f ms\n', ...
        queueLatency(s));

    fprintf('Total Latency = %.2f ms\n', ...
        totalLatency(s));

```

end

%% 15. QoS Verification

```
fprintf('\n');
fprintf('=====\n');
fprintf('QoS VERIFICATION\n');
fprintf('=====\n');
```

for s = 1:3

rateOK = throughput(idx,s) >= targetRate(s);

latencyOK = totalLatency(s) <= maxLatency(s);

if rateOK && latencyOK

fprintf('%s : QoS SATISFIED\n', ...
serviceName{s});

else

fprintf('%s : QoS NOT SATISFIED\n', ...
serviceName{s});

end

end

%% 16. Plot - Throughput vs SNR

figure;

plot(SNR_dB, throughput(:,1), '-o', ...
'LineWidth', 1.5);

hold on;

plot(SNR_dB, throughput(:,2), '-s', ...
'LineWidth', 1.5);

plot(SNR_dB, throughput(:,3), '-^', ...
'LineWidth', 1.5);

grid on;

xlabel('SNR (dB)');

ylabel('Throughput (Mbps)');

title('Throughput of Edge-Assisted Smart City Services');

legend(serviceName, ...
'Location', 'northwest');

%% 17. Plot - Resource Allocation

```
figure;  
  
bar(RB);  
  
set(gca, 'XTickLabel', serviceName);  
  
ylabel('Resource Blocks');  
  
title('5G Radio Resource Allocation');  
  
grid on;
```

%% 18. Plot - Latency Components

```
figure;  
  
latencyMatrix = [txLatency;  
                 edgeLatency;  
                 queueLatency];  
  
bar(latencyMatrix);  
  
set(gca, 'XTickLabel', serviceName);  
  
ylabel('Latency (ms)');  
  
title('Latency Components of Smart City Services');  
  
legend('Transmission', ...  
       'Edge Processing', ...  
       'Queueing', ...  
       'Location', 'northwest');  
  
grid on;
```

%% 19. Plot - Total Latency vs QoS Limit

```
figure;  
  
bar([totalLatency; maxLatency]);  
  
set(gca, 'XTickLabel', serviceName);  
  
ylabel('Latency (ms)');  
  
title('Total Latency and Maximum QoS Latency');  
  
legend('Calculated Latency', ...  
       'Maximum Allowed Latency', ...  
       'Location', 'northwest');  
  
grid on;
```

%% 20. Final Summary

```
fprintf('\n');
fprintf('=====\n');
fprintf('EDGE-ASSISTED SMART CITY SUMMARY\n');
fprintf('=====\n');
```

```
fprintf('Carrier Frequency : %.1f GHz\n', fc_GHz);
fprintf('System Bandwidth : %d MHz\n', BW_MHz);
fprintf('MIMO : %d x %d\n', Nt, Nr);
fprintf('Transmit Power : %d dBm\n', Ptx_dBm);
fprintf('Noise Power : %d dBm\n', Noise_dBm);
fprintf('Number of Users : %d\n', Nusers);
fprintf('Edge Nodes : %d\n', Nedge);
fprintf('Resource Blocks : %d\n', N_RB);
```

```
fprintf('\nResults at 15 dB SNR:\n');
```

```
for s = 1:3
```

```
    fprintf('%s:\n', serviceName{s});
```

```
    fprintf(' Throughput = %.2f Mbps\n', ...
        throughput(idx,s));
```

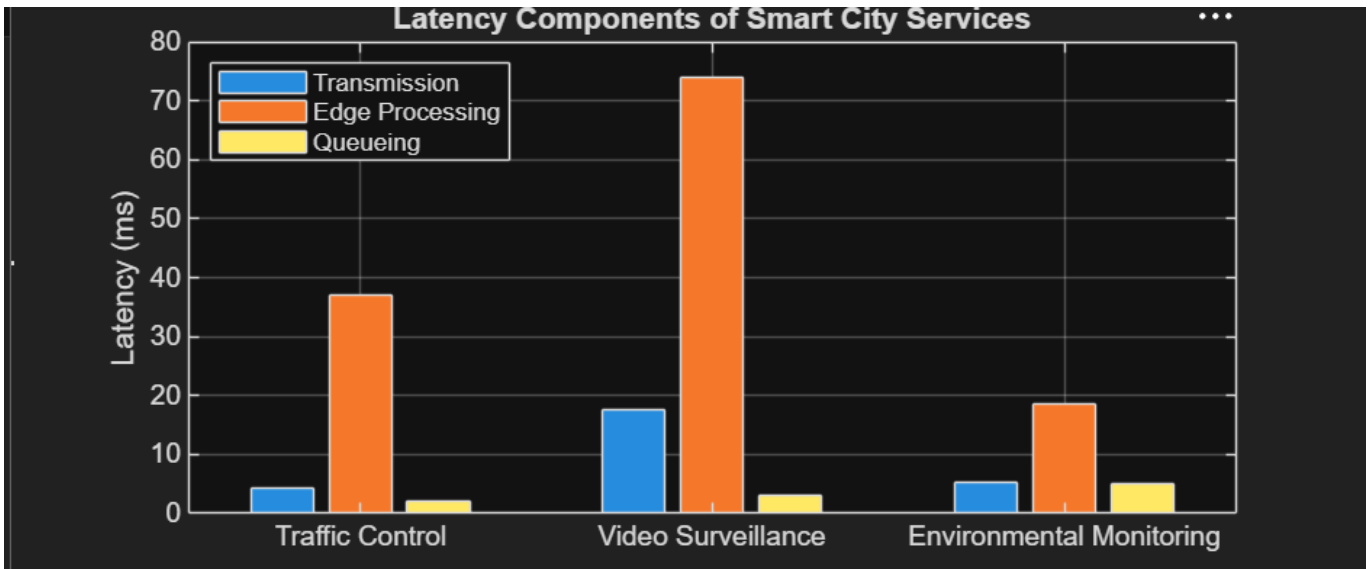
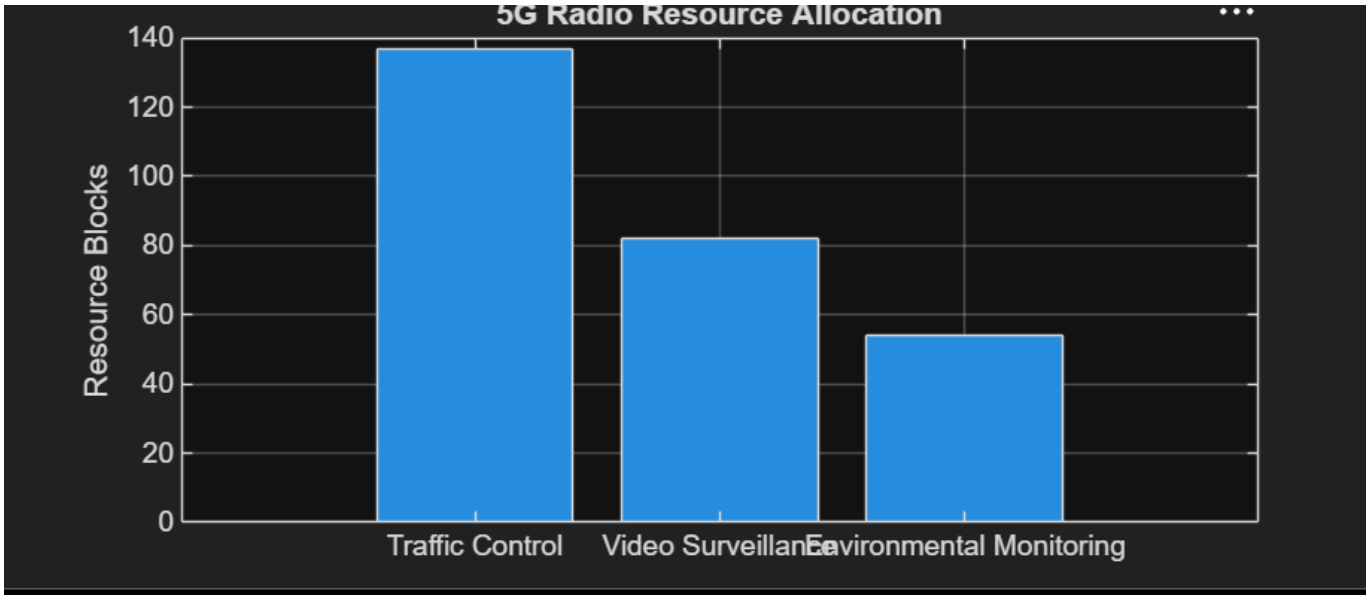
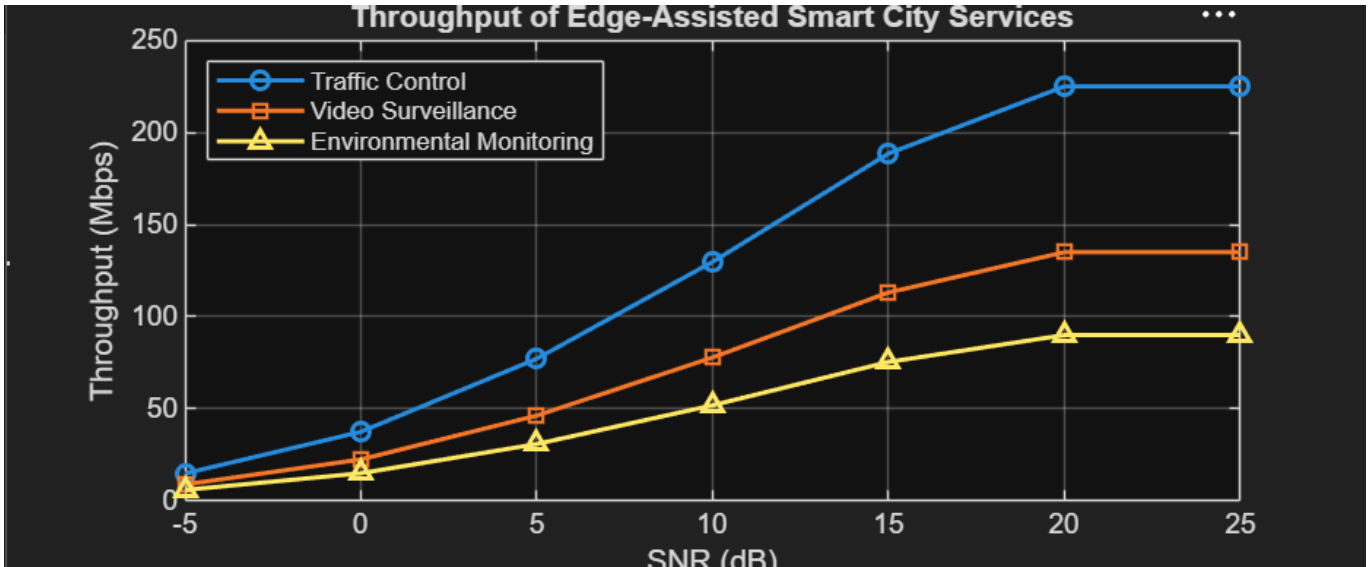
```
    fprintf(' Total Latency = %.2f ms\n', ...
        totalLatency(s));
```

```
end
```

```
fprintf('\nSimulation completed successfully.\n');
```

6. Output:

Slice	Bandwidth	Throughput at 15 dB	Minimum Rate	Rate Requirement
eMBB	50 MHz	188.54 Mbps	40 Mbps	Satisfied
URLLC	30 MHz	113.13 Mbps	15 Mbps	Satisfied
mMTC	20 MHz	75.42 Mbps	5 Mbps	Satisfied



7. Final Design Specification:

Design Item	Final Specification
Architecture	Virtualized shared RAN with QoS-aware slice orchestration
Carrier	3.5 GHz
Bandwidth	100 MHz
MIMO	2×2
Users	30
Slices	eMBB / URLLC / mMTC
Bandwidth split	50 / 30 / 20 MHz
RB weights	50 / 30 / 20 %
Scheduler	Weighted QoS-aware dynamic scheduler
Verification	MATLAB SNR-throughput and RB-allocation simulation

10. Summary:

A virtualized wireless access framework has been designed for next-generation communication using a shared 3.5 GHz, 100 MHz 2×2 MIMO radio access network. The physical resources are abstracted into a virtual resource pool and divided into eMBB, URLLC and mMTC slices with different bandwidth, resource and QoS requirements. The required formulas for SNR, spectral efficiency, throughput and resource allocation were applied to the design. MATLAB simulation verifies the relationship between SNR and slice throughput and demonstrates that the proposed allocation can satisfy the selected minimum rate requirements at the design operating point.